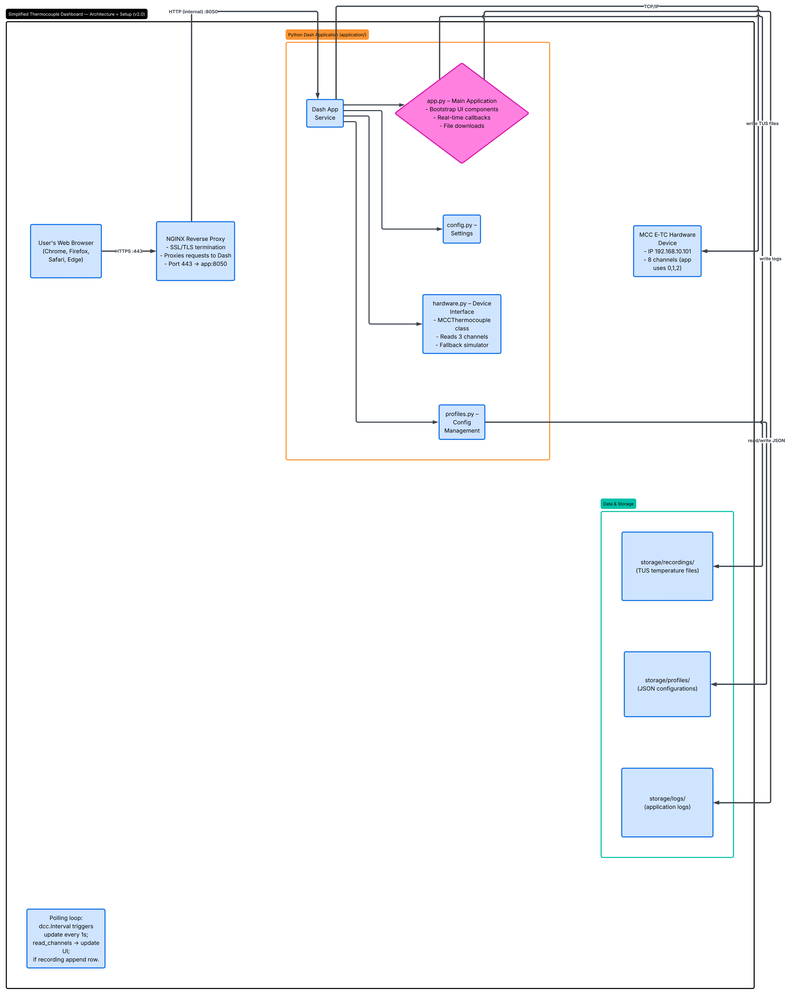

Simplified Thermocouple Dashboard — Architecture + Setup (v2.3)
HTTP (internal) :8050
TCP I/O
Python Dash Application (application/)
Dash App Service
app.py – Main Application
- Bootstrap UI components
- Real-time callbacks
- File downloads
config.py – Settings
hardware.py – Device Interface
- MCCThermocouple class
- Reads 3 channels
- Failback simulator
profiles.py – Config Management
write CSV files
write logs
read & write JSON
User's Web Browser (Chrome, Firefox, Safari, Edge)
HTTPS :443
NGINX Reverse Proxy
- SSL/TLS termination
- Proxies requests to Dash
- Port 443 → app:8050
MCC E-TC Hardware Device
- IP: 192.168.10.101
- 8 channels (app uses 0,1,2)
Data & Storage
storage/recordings/ (TUS temperature files)
storage/profiles/ (JSON configurations)
storage/logs/ (application logs)
Polling loop:
dcc.Interval triggers update every 1s
read_channels → update UI
if recording append row.


Thermocouple Dashboard – Setup Process
Thermocouple Dashboard – Setup Process
Start setup
Use Docker?
Yes
No
Docker
Prereqs: Docker + Docker Compose
docker-compose build
Using bind-mount dirs instead of volumes?
Yes
No
Create storage/recordings, storage/profiles, storage/logs
Use Docker volumes
docker-compose up
Open: http://localhost:8050
Permission error on volumes?
Yes
No
Fix ownership or run as root
Proceed
Local (Python)
Prereqs: Python 3.8+, Git
Clone repo
Create venv (.venv)
Activate venv
pip install -r requirements.txt
Use real MCC hardware?
Yes
No
Edit application/config.py: set DEVICE_IP
Use simulator (automatic fallback)
Run: python application/app.py (repo folder may be 'application')
Open: http://localhost:8050
Port 8050 in use?
Yes
No
Change port in app.py or compose
Proceed
Need HTTPS / production?
Yes
No
Setup Process (Local / Docker / HTTPS)
Generate SSL certs (self-signed ok for local): cert.pem + key.pem
Place in ./ssl/
Update docker-compose: mount cert/key into NGINX
docker-compose up -d (Deployment / Setup)
Open: https://localhost (accept warning if self-signed)
Certificate warning?
Yes
No
Accept self-signed warning
Proceed
Setup complete
Hardware test: Run 'python application/test_hardware.py'
Test hardware
Connected
Offline → simulator
Test hardware